



RISK ASSESSMENT

INCREASED CIRCULATION OF MPOX VIRUS CLADE Ib IN BELGIUM

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SIGNAL

Increased circulation of mpox virus Clade Ib/sh2023 in Belgium

DESCRIPTION

Event

Following the detection of autochthonous transmission of mpox virus (MPXV) Clade Ib/sh2023 in the EU/EEA (see [Primary Risk Assessment of 5 November 2025](#)), the number of cases of mpox due to Clade Ib has increased over the past months.

As of 6 May, **77 cases of mpox** have been reported to Sciensano with onset¹ between January and April 2026, of which 40 in April (*Annex Figure 1*). This compares to 30 cases in the period January to April 2025. Almost half of the cases in 2026 (36/77, 47%) were reported in Brussels. Among 46 cases in 2026 for which the National Reference Centre has clade information, 44 (96%) were due to Clade Ib.

Out of 53 cases in 2026 with available information, 11 (21%) were hospitalised among whom five for clinical reasons, one for isolation and five for whom the reason was not available. In 2025, there were eight hospitalisations among 55 cases with available information (15%), among whom three for clinical reasons, four for isolation and one for whom the reason was not available.

In 2025, 27 cases out of 52 cases with available information (52%) had recent travel history (n = 25) or were contacts of a case with travel history (n=2). In contrast, **only five cases in 2026** (out of 39 with information available, 13%) had **recent travel history**.

Out of 63 cases with information available, 57 (**90%**) were **men reporting sexual contacts with other men**. This is higher than in 2025, when the corresponding

¹ Throughout this document, cases are reported by date of onset unless stated otherwise. When date of onset is unknown, date of diagnosis or notification is used instead.

	figure was 36 out of 50 (72%). The most reported exposure settings remain private settings and sexual contacts at private parties, clubs, sauna or similar. Exposures at large events are reported for only a small minority of cases. The proportions of cases who use PrEP and of cases living with HIV are stable compared to 2025. Among 52 cases in 2026 with available information, 18 were PrEP users (35%; 2025: 36%). Among 46 cases in 2026 with available information, 8 were living with HIV (17%; 2025: 16%). European context In the EU/EEA, the number of cases per month is largely stable (<i>Annex Figure 2</i>). The number of cases due to Clade I has increased sharply since the autumn of 2025 before stabilising from January 2026 onward, while the number of cases due to Clade II shows a decreasing trend (<i>Annex Figure 3</i>). In March², Belgium had the highest incidence among European countries with over one million inhabitants (<i>Annex Figure 4</i>). Global context Following the 2022 epidemic due to Clade IIb/sh2017 MPXV, the burden of mpox has been the highest in countries on the African continent, as highlighted in previous risk assessments on mpox. Over the past year, a sustained decline in case numbers has been observed (Figure 5) with occasional flare-ups, most recently in Madagascar in 2026. Whereas the WHO AFRO region accounted for 80% of global cases notified in 2025, this proportion has reduced to 35% among cases notified in January and February 2026, with a concomitant increase in the number of cases notified in the WHO EURO region, from 5% to 18% (1). The Public Health Emergency of International Concern (WHO) and Public Health Emergency of Continental Security (Africa CDC) regarding the 2024 mpox upsurge were lifted respectively in September 2025 and in January 2026 (2,3).
Type of risk: UNUSUAL	The increased number of cases in Belgium in 2026 is unusual in context of the relatively stable trends (typically 0-9 cases per month) observed after 2022. This is the largest increase in cases of mpox observed since 2022 and the first upsurge of mpox due to Clade Ib/sh2023. The low proportion of cases with travel history, combined with the decreasing trend in Africa and stable trend in Europe indicates increasing transmission in Belgium, rather than an increasing number of imported cases from an external outbreak.
Impact: LOW	The severity of disease is considered as low, both for infection with Clade Ib/sh2023 and with Clade IIb/sh2017. Previous assessments relied on evidence for Clade Ib/sh2023 from the outbreak on the African continent. The co-circulation of both clades in a number of European countries has allowed for the first comparisons of the severity of Ib/sh2023 and IIb/sh2017 in the same setting (4). Overall, the rate of hospitalisation is low and slightly higher for Clade Ib/sh2023 (14%) than Clade IIb/sh2017 (7%). No deaths have been reported following infections with clade Ib/sh2023 in the EU/EEA. In case of infection, children, pregnant women and people with weak immune systems, including people living with HIV that is not controlled, are considered at higher risk for serious illness.
Exposed population	<u>At-risk population</u>

² Last month of data for EU; note substantial further increase in Belgium in April

The population most likely to be exposed to MPXV Clade Ib/sh2023 are MSM with multiple or frequently changing sexual partners. In addition, there is an increased risk for household or sexual contacts of persons with an MPXV infection, travellers to countries/regions with established circulation and healthcare workers caring for mpox cases without appropriate PPE.

For the general population in Belgium, the likelihood of exposure remains very low.

Population immunity

In the 2024 round of the European MSM and trans people internet survey (EMIS), **24% of Belgian respondents reported having been vaccinated** with two vaccine doses. Furthermore, **over 50% of Belgian respondents indicated an intention to get vaccinated if mpox cases were to increase rapidly, and around one in five respondents (over one in four unvaccinated respondents) indicated a past intention to get vaccinated which was not realised** because the vaccine was not available to them at the time (*unpublished data*). The EMIS protocol aimed at online recruitment to target the sexually active population through MSM-targeted dating apps, social media and (inter)national HIV-LGBT organisations (5). It should be noted that due to this convenience sampling approach, the percentages reported here may not be representative of the entire Belgian MSM population, nor of the population most at risk of exposure to MPXV.

Across 2022-2026, 1002 confirmed cases of mpox have been reported in Belgium (of which 790 in the epidemic year 2022). In 2025-2026 there were 146 confirmed mpox cases. **Among the 56 cases (out of 146, 38%) for which information on vaccination status is available, 11 (20%) had been previously vaccinated with two doses, and 1 (2%) with one dose.** Fifteen cases were born before 1976 (out of 136, 11%) and therefore expected to have received childhood smallpox vaccination. Three cases reported a previous infection, out of 25 with available information for this variable.

A seroprevalence of approximately 50% (Orthopoxvirus IgG) comprising natural and vaccine-induced immunity was identified among PrEP users at the Institute of Tropical Medicine (ITM) after the 2022 mpox epidemic (6). Unpublished results of research performed by ITM show a **reduction of seroprevalence to approximately 20-30% in samples collected in 2025.** However, it is unclear whether detectable antibodies are required for protection, to what degree the observed decrease is driven by waning or by entry of immune-naïve individuals, and to what extent these results are generalizable at the national level.

Likelihood of spread

HIGH

The risk of further spread among at-risk groups in Belgium is high. The expected increased rate of sexual contacts in the context of the spring and summer events seasons, including Pride events, could be a catalyst for further transmission.

An ECDC modelling study (pre-print) assessed the probability of a MPXV Clade Ib/sh2023 outbreak following the **introduction of a single undetected case** in an MSM sexual network of 10.000 individuals, representative of a metropolitan city (7). It modelled scenarios taking into account some population immunity (varying by scenario), and varying transmissibility of Clade Ib/sh2023 (equal to or 15% higher than Clade IIb/sh2017). The simulations showed that **targeting the top 15% most sexually active individuals**, even with a modest increase in vaccination uptake of 2.5 percentage points (e.g. 500 doses for 250 of the 10.000 individuals), substantially lowers risk of a large outbreak (> 100 cases). A more detailed description of the results of the simulation study is available in Annex.

PREPAREDNESS & CONTROL MEASURES ALREADY IN PLACE

Preparedness

Contact tracing

Mandatory notification and contact tracing for infection prevention and control is currently in place.

Access to testing / diagnosis

Since January 2025, testing and diagnosis capacity has been centralised within the [National Reference Centre for mpox](#) and is free of charge for the patient. Testing (at cost to patient) in other laboratories can be considered taking into account the cost of sample transport to the NRC. PCR testing at the NRC is available daily during weekdays. A properly completed '[reference form](#)' should be attached. Clade subtyping (I, II, a, b) can be performed on the same day in most cases. The NRC for mpox performs genomic sequencing (TAT 7 days) on a subset of mpox samples upon request by the Health Authorities or for research purposes.

Information about procedures for sampling and transport for clinical laboratories is available on the webpage [sampling procedures](#).

Specific Control Measures

Awareness

The autochthonous circulation of Clade Ib/sh2023 and increased case count in January-March relative to same period last year were reported in the Flash Infectious Diseases of April 2026 to raise awareness among clinicians. The Flash article also reiterated that mpox is mandatory notifiable.

Vaccination

In the revised recommendations of the Superior Health Council (SHC) on the Belgian mpox vaccination strategy (8), published in March 2026, subcutaneous vaccination (two doses of 0.5ml) is recommended for previously unvaccinated individuals in the groups at highest risks of exposure.

1. **Pre-exposure prophylaxis** is recommended to be offered to
 - a. Persons with frequently changing sexual partners.
 - b. Individuals travelling, in case of specific high-risk exposure scenario's: anticipated high-risk sexual contacts, humanitarian and health care workers after risk assessment, and others if deemed essential by an infectiologist based on medical risk factors and high risk of exposure.
2. **Post-exposure prophylaxis** is recommended to be offered to close contacts of confirmed, probable or suspected mpox cases.

Intradermal administration (two doses of 0.1ml) can be used in conditions of limited vaccine stock, in immunocompetent individuals only.

This list is provided as a high-level summary of the recommendations by the SHC. Detailed parameters for each eligible group and considerations for particular risk groups can be consulted in the advice SHC n° 9900 (8).

The vaccination strategy against mpox was discussed during the meeting of the **Risk Management Group of 5 February 2026** (9). In Flanders, doses are currently reserved for post-exposure vaccination, but pre-exposure vaccination of groups at highest risk is considered. In Brussels, vaccines are to be administered sparingly and only as first dose (but two doses for immunocompromised individuals). It is assumed that the majority of those at highest risk are already vaccinated. In Wallonia, as well as in the German-speaking Community, the vaccine doses are offered post-exposure, and – for those at high risk – as pre-exposure vaccination.

PUBLIC HEALTH IMPACT IN BELGIUM

LOW-MEDIUM

Given the high risk of exposure and low severity, the **risk for men who have sex with men with multiple or frequently changing sexual partners is considered to be medium. The risk for the general population in Belgium remains low** based on very low risk of exposure and low severity.

An important **uncertainty** is the degree to which the observed increase is driven by the start of the spring and summer event season versus higher intrinsic transmissibility of Clade Ib. Due to lack of co-circulation of Clades Ib/sh2023 and Ib/sh2017 prior to October 2025, studies directly comparing the transmissibility of the two clades in the same setting are currently not available.

RECOMMENDATIONS

1. **Contact tracing** should be maintained as an essential approach to limit spread.
2. **Surveillance** should be maintained and strengthened where necessary
 - **Data quality** of mandatory notifications should be improved, in particular for variables concerning immunity status (previous mpox infection or vaccination) and exposure setting.
 - For probable cases with very clear clinical presentation, the need to confirm infection with MPXV via PCR can be evaluated on an individual basis. However, all suspected, probable and confirmed cases should continue to be reported via mandatory notifications.
 - Clade identification should be maintained for all positive samples
 - **Sequencing** of selected samples can be performed to inform epidemiological and virological research, in particular with respect to the characteristics of Clade Ib/sh2023.
3. **Risk communication** should be initiated as soon as possible to inform the at-risk population of the current epidemiology, risk and preventive measures, in collaboration with community-based organisations.
4. Implement **vaccination** in accordance with the recommendations of the Superior Health Council.
 - There is need for a clear operational translation of the recommendations by the Superior Health Council, ideally harmonised across the regional health authorities
 - Within the eligible population, targeted pre-exposure vaccination of the most sexually active individuals has the strongest limiting impact on the spread of infection and subsequent outbreaks.
 - To reduce transmission in the at-risk population, pre-exposure vaccination is considered more effective than post-exposure vaccination. The latter can prevent disease in the exposed individual but has limited impact on circulation of the virus.
 - In order to inform the required degree of prioritisation, transparency about the stocks available for prevention is important.
 - Intradermal vaccination is indicated in conditions of limited vaccine stocks, for immunocompetent individuals. If intradermal administration would be offered, planning is required to ensure that a sufficient number of doses can be administered to make full use of each vial. It is therefore more suitable for application in the context of (planned) pre-exposure vaccination than (reactive) post-exposure vaccination.

- Post-exposure vaccination can be initiated before laboratory confirmation in index case when there is a strong suspicion of infection in the index.

5. In general, there is a need for a **coherent, long-term strategy** for management of mpox as an endemic disease in Belgium, covering sensibilisation, contact tracing, vaccination, and surveillance.

ACTIONS

Sciensano

- Coordinate further efforts to improve data quality with Federated Health Authorities
- Resume monthly epidemiological bulletins
- Communicate to laboratories to remind of testing modalities including analysis request forms (and reference form to be attached) and sample transport.

National Reference Centre for Mpox

- Maintain clade subtyping for molecular surveillance

Federated Health Authorities

- Improve reporting of immunity status and exposure setting

Federal and federated Health Authorities

- Communicate to general practitioners emergency physicians, gynaecologists, dermatologists to further raise awareness of the increased circulation of mpox, the requirement to report suspected cases via mandatory notifications, and to advice on where to refer suspected cases (for testing) and close contacts (for post-exposure vaccination).
- Coordinate promptly with community-based organisations to increase risk communication and prevention efforts, with particular attention for sexual contacts in the context of large events in the upcoming months.
- Taking into account the current epidemiological evolution, review and if necessary adapt the vaccination strategy. Communicate vaccination strategy clearly to stakeholders.

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ANNEXES

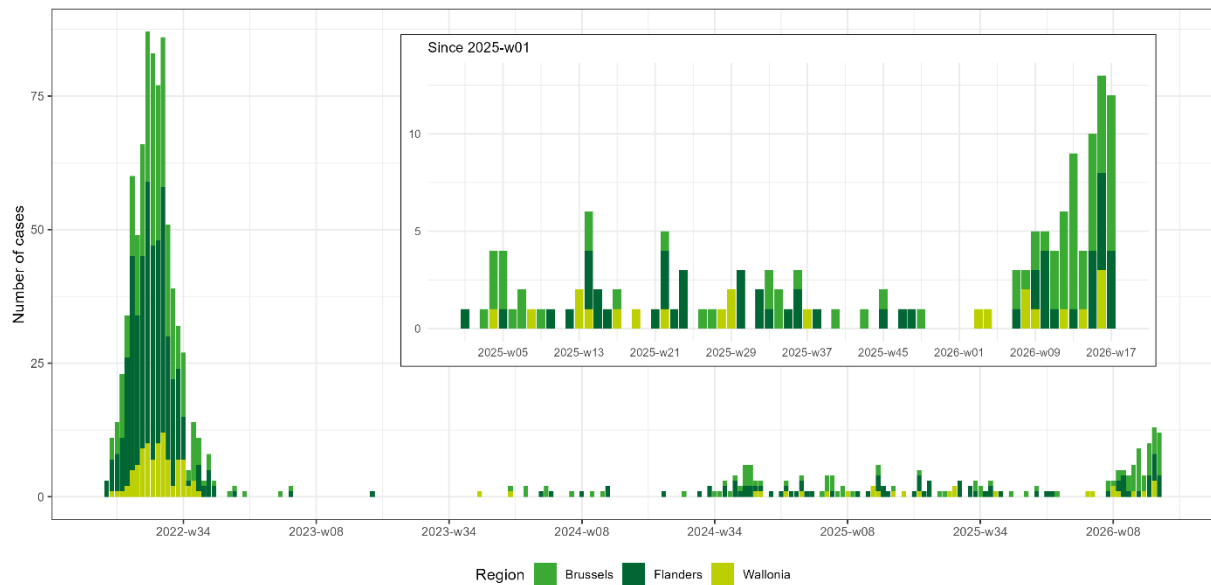


Figure 1: Mpxv cases by region since May 2022, Belgium. Source: Sciensano (mandatory notifications)

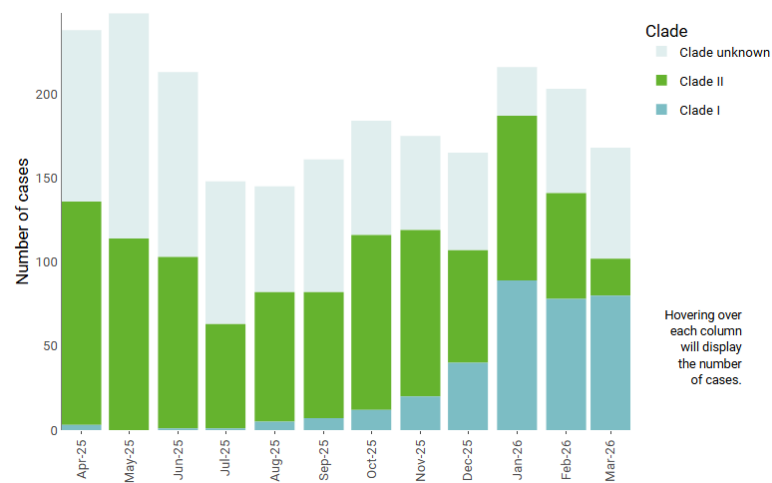


Figure 2: Mpxv cases by clade in the past 12 months (April 2025 to March 2026), EU/EEA. Source: ECDC

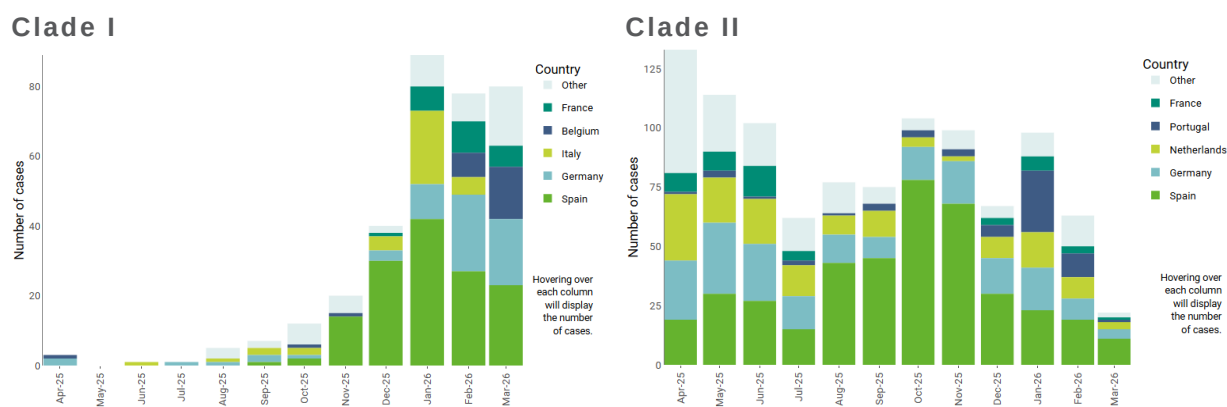


Figure 3: Mpox cases by clade and country in the past 12 months (April 2025 to March 2026), EU/EEA. Source: ECDC

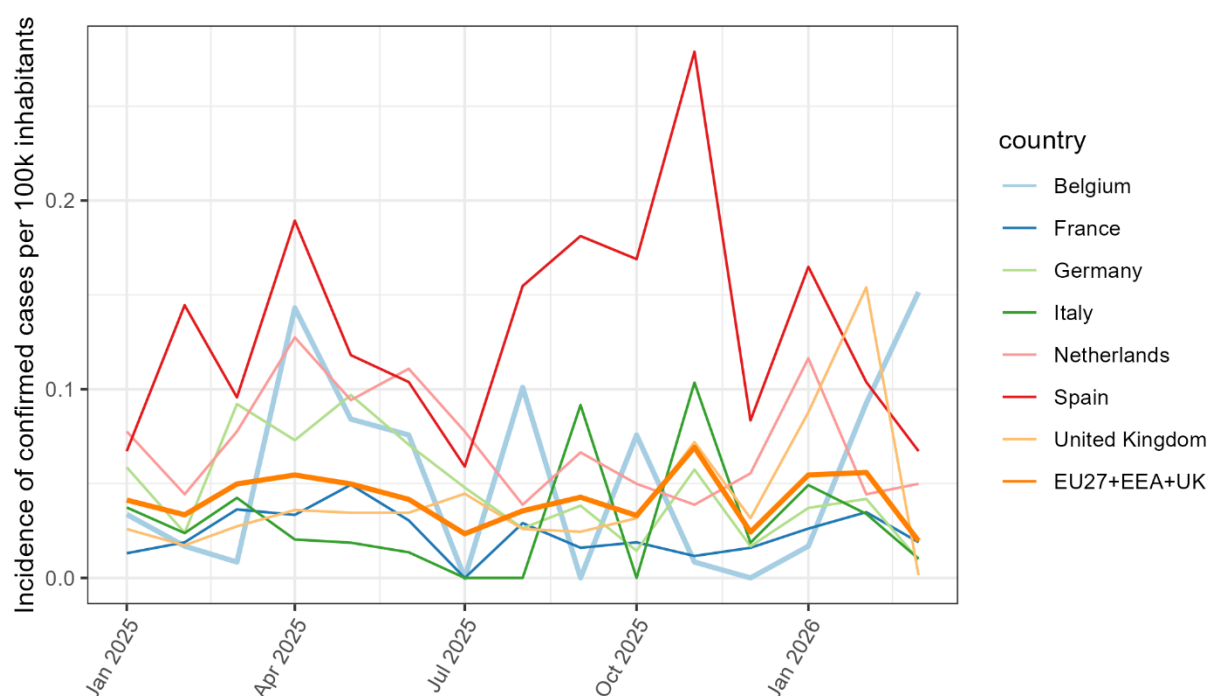


Figure 4: Incidence of confirmed cases of mpox (all clades) per 100k inhabitants (January 2025 to March 2026), selected countries (EU/EAA and UK) based on highest number of cases and/or incidence. Increase in April 2026 in Belgium not included in data; note that the last month of data is subject to reporting delay. Source: WHO

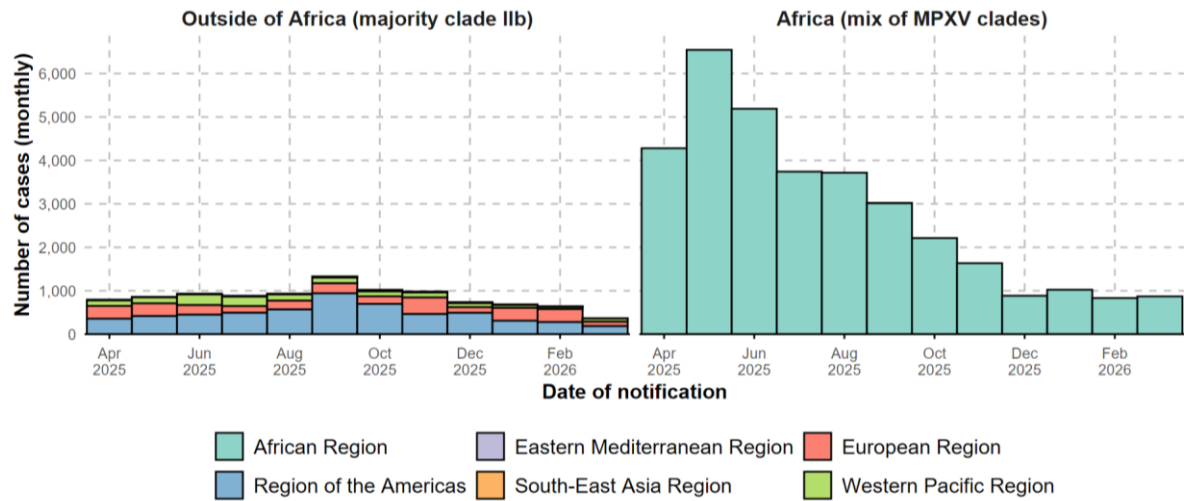


Figure 5: Trends in global mpox cases by WHO region. Data as of 31 March 2026; note that the last month of data is subject to reporting delay. Source: WHO.

Summary of main points of the pre-print article “Modelling the outbreak risk of monkeypox virus Clade Ib among MSM in the EU/EEA and the impact of targeted vaccination” (7)

Impact of (targeted) vaccination

The impact of targeted vaccination on small (>10 cases) and large outbreaks (>100 cases) was analysed with different scenarios: an increased vaccination uptake of 2,5%pt (low) and 5%pt (high) with different degrees of targeting towards the most sexually active people: top 15% (highly targeted), 25% (medium) and 50% (low) (Figure 6).

The high additional uptake scenario (+5%pt) leads to an important reduction of the risk of both small (left) and large (right) outbreaks, while the low additional uptake scenario (+2.5%pt) leads to a substantial reduction of the risk of large outbreaks but a more limited reduction of the risk of small outbreaks.

In both scenarios of low (+2.5%pt) or high additional uptake (+5%pt), and for both small and large outbreaks, **targeting the top 15% most sexually active individuals is provides a substantial additional reduction of outbreak risk (cfr. purple) compared to a broader uptake among less targeted group of the population (green).** Strikingly, in these simulations a highly targeted approach (top 15%) combined with low additional uptake (+2.5%pt) leads larger reduction of outbreak risk (both small and large) than an approach with higher additional uptake (+5%pt) and a lower degree of targeting (top 50%).

It should be noted that these simulations address scenarios following initial introduction of a single case and thus do not correspond directly to the Belgian context.

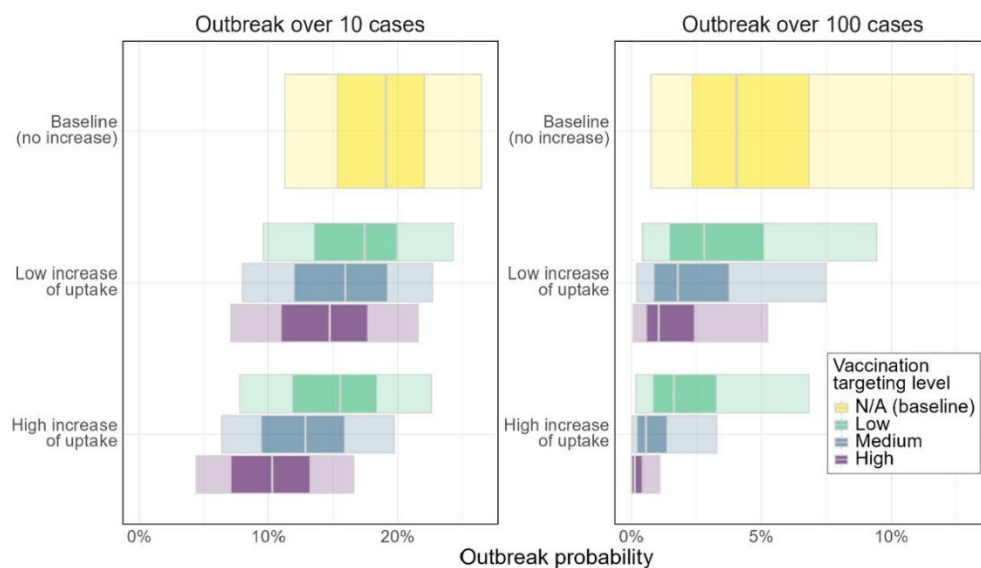


Figure 6: Probability of small or large outbreaks from a singular imported case, depending on vaccination uptake among different targeted sexually active groups. Dark and light bars indicate the 50% and 95% confidence intervals. Note the different x-axes between the two plots.

DESCRIPTION METHODOLOGY	
EVIDENCE	
AVAILABILITY OF EVIDENCE	The sources used for this risk assessment have been duly referenced. As indicated in the text on 'public health impact' several currently unknown factors could influence the potential public health impact. The main elements of uncertainty are relative transmissibility of Clade Ib/sh2023 compared to Clade Ib/sh2017 and the level of immunity in the at-risk population in Belgium.
EXPERTS	
SELECTION PROCESS	Experts that have been involved in previous risk assessments on the same topic were invited to contribute to this RA. They were selected based on relevant expertise. Additionally, permanent RAG members (as defined by the protocol agreement) were involved. The full list of RAG experts is available at https://www.sciensano.be/en/rag-experts.
EXPERT LIST	Leïla Belkhir (St Luc UCL), Valentin Blaison (ExAequo), Koen Blot (Sciensano), Isabel Brosius (ITM), Steven Callens (UZ Gent/NITAG), Steven De Keukeleire (RZ Tienen), Marie-Angélique De Scheerder (UZ Gent), Edwinne Deprez (Centre S), Pierre-Louis Deudon (FAMGB/CMG), Wouter Dhaeze (DZ), Achille Djena (AVIQ), Veerle Doossche (Sensoa), Jean-Cristophe Goffard (UT St Pierre), Naima Hammami (DZ), Niel Hens (U Hasselt/U Antwerp), Bart Hoorelbeke (FOD), Amaryl Lecompte (Sciensano), Laurens Liesenborghs (ITM), Agnes Libois (CHU St Pierre), Christelle Meuris (CHR Verviers), Geert Molenberghs (U Hasselt/KU Leuven), Annick Paeps (DZ), Sophie Quoilin (CHU St Pierre), Florence Rolin (Vivalis), Anna Schmelz (GSC), Jorgen Stassijns (RAG/Sciensano), Stefan Teughels (Domus Medica), Heidi Theeten (DZ), Marjan Van Esbroeck (ITM), Koen Vercauteren (ITM), Anne Julie Verhaege (DZ), Emma Van de Voorde (Violet), Liesbeth Aertsens (Violet), Martine Claeyssens (Violet), Joelle Konings (DZ), Sophie Godenne (Alias), Thierry Martin (Plateforme Prevention Sida)
DECLARATION CONFLICT OF INTEREST	All experts that have submitted a DOI have been assessed by the RAG coordination team. No relevant conflicts were disclosed. An ad-hoc DOI will be requested from those experts who did not yet submit a DOI.
CONSULTATION PROCESS	
MEETING	Draft RA sent on 11/5/2026 Online expert meeting on 13/5/2026 Final draft approved via mail on 20/05/2026 Final RA published on 26/5/2026
SURVEY	No survey
DECISION PROCESS	
CONSENSUS	Consensus on the final version on 20/05/2026
OBJECTIONS	None